

Pinpoint Tests — OSC Equipped Vehicle

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)	
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

When the electrical connector or solenoid body is disconnected, inspect the connector for terminal condition, corrosion and contamination. Also inspect the connector seal for damage. Clean, repair or install new components as required.

Power is routed through the Transmission Solenoid Power Control (TSPC) relay (inside the PCM) to all transmission solenoids. If the power circuit to the transmission solenoids or the TSPC relay fails open, then all solenoids are failed electrically OFF. CHECK for open, short to ground or the transmission connector disconnected. The TSPC relay will disable power to the transmission solenoids when certain transmission DTCs are set.

Shift Solenoid Pre-Diagnosis

Solenoid Operation Chart

Selector Lever Position	PCM Commanded Gear	Shift Solenoid					TCC NL
		SSA NL (1,2,3,4)	SSB NH (3,5,R)	SSC NL (CB 2,6)	SSD NH (CB L,R/C 4,5,6)	SSE NC	
P	P	Off	On	Off	Off	Off	Off
R	R	Off	Off	Off	Off	Off	Off
N	N	Off	On	Off	Off ^a	Off	Off
D	1	On	On	Off	Off ^b	Off	Off
	2	On	On	On	On	Off	Off
	3	On	Off	Off	On	On	On/Off
	4	On	On	Off	Off	On	On/Off
	5	Off	Off	Off	Off	On	On/Off
	6	Off	On	On	Off	On	On/Off
M ^c	—	—	—	—	—	—	—
2	2	On	On	On	On	Off	Off

Selector Lever Position	PCM Commanded Gear	Shift Solenoid					TCC NL
		SSA NL (1,2,3,4)	SSB NH (3,5,R)	SSC NL (CB 2,6)	SSD NH (CB L,R/C 4,5,6)	SSE NC	
1	1	On	On	Off	Off	On	Off

CB = Clutch brake

NC = Normally closed

NH = Normally high

NL = Normally low

^aSolenoid state will change if vehicle is moving forward with the selector lever in the NEUTRAL position.

^bSolenoid is On when vehicle is above 3 mph.

^cAll gears available upon driver request.

Pinpoint Tests

Refer to Wiring Diagrams Cell [30](#) for schematic and connector information.

NOTE: Refer to all TSBs and On-Line Automotive Service Information System (OASIS) messages that pertain to the transmission concern and follow the procedure as described.

NOTE: Vehicles equipped with a 3.7L engine use PCM [C175T](#).

NOTE: Vehicles equipped with a 5.0L or 6.2L engine use PCM [C1381T](#).

NOTE: Vehicles equipped with a 3.5L engine use PCM [C1551E](#).

PINPOINT TEST A : TRANSMISSION CONTROL SOLENOIDS

NOTE: Refer to the Transmission Vehicle Harness Connector illustration within the Transmission Connector Layouts procedure.
NOTE: Refer to the Transmission Leadframe Connector illustration within the Transmission Connector Layouts procedure.
NOTE: If the scan tool is unable to access the transmission PIDs, refer to Powertrain Control/Emissions Diagnosis (PC/ED) manual.
NOTE: Read and record all DTCs.
A1 CHECK THE SOLENOID POWER CIRCUIT FOR VOLTAGE
NOTE: The <u>PCM</u> removes power from the Transmission Solenoid Power Control (TSPC) circuit when the key is on and the transmission harness is disconnected. Use the meter’s min/max function to record the voltage before the <u>PCM</u> removes power. - Ignition OFF. - Disconnect: Transmission C1575. - Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.

- **NOTE:** Use a meter with a min/max feature and/or recording capabilities to obtain accurate measurements.

Set the meter to measure DC voltage.

- Set the meter range to greater than 15 volts.
- Enable the min/max or recording feature.
- **NOTE:** Make sure all test connections are made at the transmission harness connector before the ignition key is turned to the ON position.

Connect the meter to [C1575](#) Pin 7 and ground.

- Ignition ON.
- Measure:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C1575 Pin 7		Ground

Is the maximum recorded voltage greater than 11 volts?

Yes	GO to A4 .
No	GO to A2 .

A2 CHECK THE SOLENOID POWER CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: PCM [C175T](#) (3.7L Engine) .
- Disconnect: PCM [C1381T](#) (5.0L and 6.2L Engines) .
- Disconnect: PCM [C1551E](#) (3.5L Gasoline Turbocharged Direct Injection (GTDI) Engine) .
- Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.
- Vehicle equipped with 3.7L engine, measure:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C175E Pin 80	Ω	C1575 Pin 7

- Vehicle equipped with 5.0L or 6.2L engine, measure:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C1381T Pin 50	Ω	C1575 Pin 7

- Vehicle equipped with 3.5L engine, measure:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C1551E Pin 80	Ω	C1575 Pin 7

Is the resistance less than 3 ohms?

Yes	GO to A3 .
No	REPAIR transmission vehicle harness circuit for an open.

A3 CHECK THE SOLENOID POWER CIRCUIT FOR A SHORT TO GROUND

- Vehicle equipped with 3.7L engine, measure:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C175E Pin 80	Ω	Ground

- Vehicle equipped with 5.0L or 6.2L engine, measure:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C1381T Pin 50	Ω	Ground

- Vehicle equipped with 3.5L engine, measure:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C1551E Pin 80	Ω	Ground

Is the resistance greater than 10,000 ohms?

Yes	INSTALL a new PCM , REFER to Section 303-14. PROGRAM the PCM with the latest calibration level. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy.
No	REPAIR transmission vehicle harness circuit for a short to ground.

A4 CHECK SOLENOID CONTROL CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: PCM [C175T](#) (3.7L Engine) .
- Disconnect: PCM [C1381T](#) (5.0L and 6.2L Engines) .
- Disconnect: PCM [C1551E](#) (3.5L GTDI Engine) .
- Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.
- Vehicles equipped with a 3.7L engine, measure the resistance of the solenoid control circuits using the following chart:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C175T Pin 45	Ω	C1575 Pin 9
C175T Pin 43	Ω	C1575 Pin 14
C175T Pin 44	Ω	C1575 Pin 13
C175T Pin 10	Ω	C1575 Pin 10
C175T Pin 11	Ω	C1575 Pin 8
C175T Pin 34	Ω	C1575 Pin 16
C175T Pin 46	Ω	C1575 Pin 3

- Vehicles equipped with a 5.0L or 6.2L engine, measure the resistance of the solenoid control circuits using the following chart:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C1381T Pin 43	Ω	C1575 Pin 9
C1381T Pin 44	Ω	C1575 Pin 14
C1381T Pin 45	Ω	C1575 Pin 13
C1381T Pin 10	Ω	C1575 Pin 10
C1381T Pin 11	Ω	C1575 Pin 8
C1381T Pin 34	Ω	C1575 Pin 16
C1381T Pin 46	Ω	C1575 Pin 3

- Vehicles equipped with a 3.5L engine, measure the resistance of the solenoid control circuits using the following chart.

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C1551E Pin 60	Ω	C1575 Pin 9
C1551E Pin 45	Ω	C1575 Pin 14
C1551E Pin 75	Ω	C1575 Pin 13
C1551E Pin 30	Ω	C1575 Pin 10
C1551E Pin 2	Ω	C1575 Pin 8
C1551E Pin 59	Ω	C1575 Pin 16
C1551E Pin 44	Ω	C1575 Pin 3

Is the resistance less than 3 ohms on the suspect circuit?

Yes	GO to A5 .
No	REPAIR the transmission vehicle harness circuit. CLEAR the DTCs .

A5 CHECK SOLENOID CONTROL CIRCUITS FOR A SHORT TO GROUND

- Measure the resistance between the transmission vehicle harness [C1575](#), harness side and ground using the following chart:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C1575 Pin 7	Ω	Ground
C1575 Pin 9	Ω	Ground
C1575 Pin 14	Ω	Ground
C1575 Pin 13	Ω	Ground
C1575 Pin 10	Ω	Ground
C1575 Pin 8	Ω	Ground
C1575 Pin 16	Ω	Ground
C1575 Pin 3	Ω	Ground

Is the resistance greater than 10,000 ohms on the suspect circuit?

Yes	GO to A6 .
No	REPAIR the transmission vehicle harness circuit. CLEAR the DTCs .

A6 CHECK SOLENOID CONTROL CIRCUITS FOR A SHORT TO POWER

- Ignition ON.

- Measure the voltage between the transmission vehicle harness [C1575](#), harness side and ground using the following chart:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
C1575 Pin 9		Ground
C1575 Pin 14		Ground
C1575 Pin 13		Ground
C1575 Pin 10		Ground
C1575 Pin 8		Ground
C1575 Pin 16		Ground
C1575 Pin 3		Ground

Is any voltage present?

Yes	REPAIR the circuit for a short to power. CLEAR the DTCs .
No	GO to A7 .

A7 CHECK FOR FLUID CONTAMINATION

- REMOVE the transmission fluid pan. REFER to Transmission Fluid Drain and Refill.
- Inspect for fine metallic contamination at the molded leadframe to solenoid connections.

Is visible metallic contamination present?

Yes	REMOVE and discard the foam strip if equipped. REMOVE and CLEAN the molded lead frame and solenoid terminals with clean transmission fluid. GO to A8
No	GO to A8

A8 CHECK MOLDED LEADFRAME RESISTANCE

- Measure resistance of the suspect solenoid circuit between the transmission harness connector [C1575](#), **component side**, and the transmission molded leadframe using the table below:

[Click to display connectors](#)

Positive Lead	Measurement / Action	Negative Lead
Transmission leadframe connector, C1575 Pin 9	Ω	Transmission molded leadframe, pin 9
Transmission leadframe connector, C1575 Pin 14	Ω	Transmission molded leadframe, pin 14
Transmission leadframe connector, C1575 Pin 13	Ω	Transmission molded leadframe, pin 13
	Ω	

Positive Lead	Measurement / Action	Negative Lead
Transmission leadframe connector, C1575 Pin 10		Transmission molded leadframe, pin 10
Transmission leadframe connector, C1575 Pin 8	Ω	Transmission molded leadframe, pin 8
Transmission leadframe connector, C1575 Pin 16	Ω	Transmission molded leadframe, pin 16
Transmission leadframe connector, C1575 Pin 3	Ω	Transmission molded leadframe, pin 3

Is the resistance less than 3 ohms on the suspect circuit?

Yes	GO to A9 .
No	INSTALL a new molded leadframe. INSTALL new foam strip (Service Part Number FL3Z-14B167-A) between molded leadframe and the solenoids. REFER to Shift Solenoids (SS).

A9 CHECK SOLENOID RESISTANCE

- Measure and record the resistance of the suspected solenoid. Shift Solenoids (SS). Compare the recorded value with the chart below.

Solenoid	Ohms
SSA	4.8 to 5.6
SSB	4.8 to 5.6
SSC	4.8 to 5.6
SSD	4.8 to 5.6
SSE	17.5 to 18.6
LPC solenoid	4.8 to 5.6
TCC solenoid	4.8 to 5.6

Is the solenoid resistance within specification?

Yes	The system is operating correctly at this time. INSTALL new foam strip (Service Part Number FL3Z-14B167-A) between molded leadframe and the solenoids. REFER to Shift Solenoids (SS)
No	INSTALL a new solenoid for the solenoid that failed the resistance check. INSTALL new foam strip (Service Part Number FL3Z-14B167-A) between molded leadframe and the solenoids. REFER to Shift Solenoids (SS).

NOTE: Refer to the Transmission Vehicle Harness Connector illustration within the Transmission Connector Layouts procedure.

NOTE: Refer to the Transmission Leadframe Connector illustration within the Transmission Connector Layouts procedure.

NOTE: If the scan tool is unable to access the transmission PIDs, refer to Powertrain Control/Emissions Diagnosis (PC/ED) manual.

NOTE: Read and record all DTCs.

B1 CHECK TFT INPUT SIGNAL

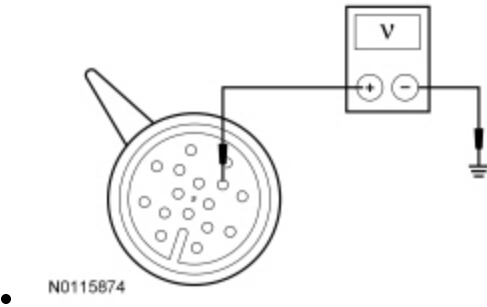
- Ignition OFF.
- Disconnect: Transmission Vehicle Harness [C1575](#).
- Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.
- Connect the scan tool.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: TFT PIDs: TFT and TFTV .

Does the TFT PID display -40°C (-40°F) and the TFTV PID display 4.96 to 5.10 volts?

Yes	GO to B2 .
No	REPAIR vehicle harness circuit for a short to ground. CLEAR the DTCs.

B2 CHECK TFT SIGNAL INPUT CIRCUIT FOR VOLTAGE

- Measure the voltage between the transmission vehicle harness [C1575](#) Pin 6, circuit, harness side and ground.

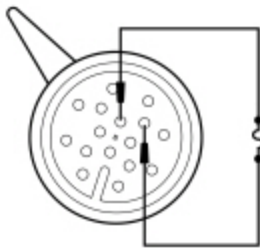


Is the voltage between 4.8 and 5.1 volts?

Yes	GO to B3 .
No	INSPECT and REPAIR transmission vehicle harness circuit VET27 (BN/YE) for an open. If an open circuit is not found, REPLACE the PCM, REFER to Section 303-14. PROGRAM the PCM with the latest calibration level. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. TEST the system for normal operation.

B3 CHECK TFT SIGNAL RETURN CIRCUIT

- Connect a fused jumper between the transmission vehicle harness [C1575](#) Pin 5 and [C1575](#) Pin 6, harness side.



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Does the TFT PID display 190°-195°C (374°-383°F) and the TFTV PID display 0 volts?

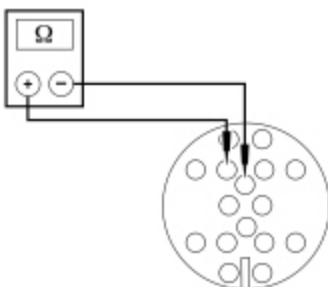
Yes	GO to B4 .
No	INSPECT and REPAIR the transmission vehicle harness signal return circuit for an open. If an open circuit is not found, REPLACE the PCM, REFER to Section 303-14. PROGRAM the PCM with the latest calibration level. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. TEST the system for normal operation.

B4 CHECK TFT SENSOR RESISTANCE

- Measure the resistance between the transmission molded leadframe [C1575](#) Pin 5 and [C1575](#) Pin 6, component side using the following chart.

Transmission Fluid Temperature (TFT)

°C	°F	Resistance
-40 to -20	-40 to -4	967K-284K
-19 to -1	-3 to 31	284K-100K
0-20	32-68	100K-37K
21-40	69-104	37K-16K
41-70	105-158	16K-5K
71-90	159-194	5K-2.7K
91-110	195-230	2.7K-1.5K
111-130	231-266	1.5K-0.8K
131-150	267-302	0.8K-0.54K



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Does the temperature to resistance specifications match?

Yes	REFER to Diagnosis By Symptom in this section to diagnose an overheating concern.
No	INSPECT and CLEAN the TTR sensor on the molded leadframe for metallic contamination. INSTALL a new molded leadframe if no debris was found, REFER to Shift Solenoids (SS) in this section. CLEAR the DTCs. TEST the system for normal operation.

Pinpoint Test C: DTCs P705, P706, P707, P708, P709, P1702, P1921

Refer to Wiring Diagrams Cell 30 for schematic and connector information.

Normal Operation

The TTR sensor has a set of Hall-effect sensors that have a pattern of ON/OFF states which are dependent on the PARK, REVERSE, NEUTRAL, DRIVE, 3, 2 or 1 position of the manual valve. The TTR sensor has digital sensor (hall effect) that converts the shifter position into a bit pattern, and a micro controller that converts the 4 bit signal into a 125 Hz pulse width modulated signal. The TTR sensor also provides signals for the starting system and the reverse lights. The PCM uses the TTR sensor signal as an input for its strategy for shifting and TCC operation. The PCM also uses the TTR sensor signal for transmission fault detection and diagnostics. The TTR sensor is integral to the leadframe. DTC s P0705, P0706, P0707, P0708, P0709, P1702 and P1921 cannot be set by an incorrectly adjusted selector lever cable.

DTC Description	Fault Trigger Condition
P0705 — Transmission Range Sensor "A" Circuit (PRNDL Input) sensor	This DTC sets with DTC P1702. The PCM sets this DTC when it has detected the TTR sensor signal duty cycle is within range but does not match the calibrated bit pattern.
P0706 — Transmission Range Sensor "A" Circuit Range/Performance	This DTC sets with DTC P1702. The PCM sets this DTC when it has detected a TTR sensor signal duty cycle that is out of range.
P0707 — Transmission Range Sensor "A" Circuit Low	This DTC sets with DTC P1702. The PCM detected a TTR sensor signal duty cycle is out of range.
P0708 — Transmission Range Sensor "A" Circuit High	This DTC sets with DTC P1702. The PCM detected a TTR sensor signal duty cycle is out of range.
P0709 — Transmission Range Sensor "A" Circuit Intermittent	This DTC sets with DTC P1702. The PCM detected a TTR sensor signal duty cycle is not out of range but is invalid.
P1702 — Transmission Range Sensor Circuit Intermittent	This DTC sets with P0705, P0706, P0707, P0708 or P0709, service P1702 last. The PCM detected that the TTR sensor signal duty cycle is not out of range but is invalid.
P1921 — Transmission Range Signal	ThisDTC sets with P0705, P0706, P0707, P0708 or P0709, service P1921 last. The PCM detected that the TTR sensor signal duty cycle is out of range.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- TTR sensor/Leadframe
- PCM

PINPOINT TEST C : DTCS P705, P706, P707, P708, P709, P1702, P1921

NOTE: Refer to the Transmission Vehicle Harness Connector illustration within the Transmission Connector Layouts procedure in this section.

NOTE: Refer to the Transmission Leadframe Connector illustration within the Transmission Connector Layouts procedure in this section.

NOTE: If the scan tool is unable to access the transmission PIDs, refer to Powertrain Control/Emissions Diagnosis (PC/ED) manual.

NOTE: Read and record all DTCs.

C1 VERIFY DTCS

- Ignition OFF.
- Select PARK.
- Connect the scan tool.
- Carry out the Key ON Engine OFF (KOEO) then the Key ON Engine Running (KOER) test.

Are Transmission Range (TR) sensor DTCs present?

Yes	GO to C2 .
No	If no DTCs are present, REFER to Diagnosis By Symptom to diagnose any symptoms that may be present.

C2 CHECK TR SENSOR ELECTRICAL OPERATION

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: TR PIDs: TR_DC and TR_FREQ .
- Move the selector lever into each range and stop.
- Observe the following PIDs: TR_DC and TR_FREQ while wiggling the harness or driving the vehicle.
- Compare the PID: TR_DC to the selector lever position duty cycle, using the following chart.

Selector Lever Position	Range (% Duty Cycle)
P	8.0 — 25.8
R	30.50 — 39.31
N	40.54 — 52.49
D	54.35 — 71.15
M	65.15 — 82
2	70.55 — 85.61
1	75.25 — 88.91

Is the TR_FREQ between 100 to 150 Hz?

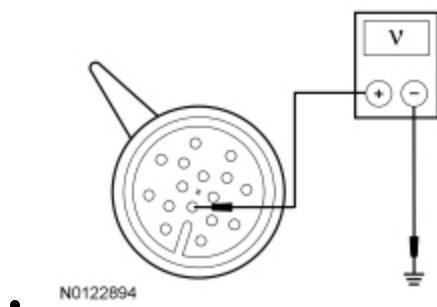
Is the TR_DC within range?

Does the TR_DC and TR_FREQ remain steady when the harness is wiggled or the vehicle driven?

Yes	The problem is not in the TR sensor system. Concern may be intermittent, INSPECT and REPAIR any loose connectors or backed out terminals. If OK, REFER to Diagnosis By Symptom in this section for further diagnosis.
No	GO to C3 .

C3 CHECK TR SENSOR VPWR CIRCUIT FOR POWER

- Ignition OFF.
- Disconnect: Transmission Vehicle Harness [C1575](#).
- Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.
- Ignition ON.
- Measure the voltage between the transmission vehicle harness [C1575](#) Pin 12, harness side and ground.

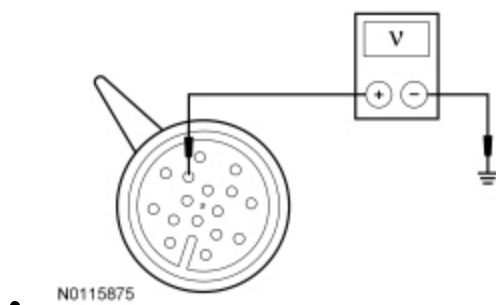


Is the voltage greater than 11 volts?

Yes	GO to C4 .
No	INSPECT and REPAIR for an open or short to ground. If a circuit fault is not found, INSTALL a new PCM, REFER to Section 303-14. PROGRAM the PCM with the latest calibration level. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. CLEAR the DTCs. TEST the system for normal operation.

C4 CHECK TR SENSOR SIGNAL CIRCUIT

- Measure the voltage between the transmission vehicle harness [C1575](#) Pin 4, harness side and ground.

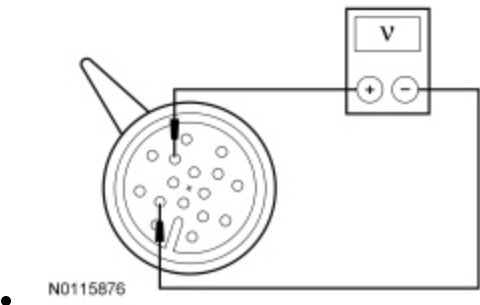


Is the voltage greater than 4.8 volts?

Yes	GO to C5 .
No	INSPECT and REPAIR vehicle harness for an open. If an open circuit is not found, INSTALL a new PCM, REFER to Section 303-14. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. CLEAR the DTCs. TEST the system for normal operation.

C5 CHECK TR SENSOR GROUND CIRCUIT

- Measure the voltage between the transmission vehicle harness [C1575](#) Pin 4 and [C1575](#) Pin 11.



Is the voltage greater than 4.8 volts?

Yes	GO to C6 .
No	INSPECT and REPAIR transmission vehicle harness for an open. If an open circuit is not found, INSTALL a new PCM, REFER to Section 303-14. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. CLEAR the DTCs. TEST the system for normal operation.

C6 CHECK TR SENSOR TO MANUAL VALVE

- Drain the transmission fluid and remove the transmission fluid pan. Refer to Fluid Pan, Gasket and Filter in this section.
- Check the connection between the TR sensor and the manual control valve. Refer to Transmission in this section.
- Inspect the molded leadframe for metallic contamination, refer to Shift Solenoids (SS).

Is the TR sensor connected to the manual valve and the molded leadframe free of debris?

Yes	INSTALL a new <u>TR</u> sensor, REFER to Shift Solenoids (SS) in this section. TEST the system for normal operation.
No	CONNECT the <u>TR</u> sensor to the manual valve, REFER to Solenoid Body Strategy in this section. CLEAN all debris from the molded leadframe. RUN the Key ON Engine OFF (KOEO) self-test. If DTCs return, INSTALL a new <u>TR</u> sensor/leadframe, REFER to Solenoid Body Strategy in this section.

PINPOINT TEST D : TSS SENSOR

NOTE: Refer to the Transmission Vehicle Harness Connector illustration within the Transmission Connector Layouts procedure in this section.

NOTE: Refer to the Transmission Leadframe Connector illustration within the Transmission Connector Layouts procedure in this section.

NOTE: If the scan tool is unable to access the transmission PIDs, refer to Powertrain Control/Emissions Diagnosis (PC/ED) manual.

NOTE: Always drive the vehicle in a safe manner according to driving conditions and obey all traffic laws.

NOTE: Read and record all DTCs.

NOTE: If DTCs P0705, P0706, P0707, P0708, P0709, P1702, P1705 and/or P1921 are set, diagnose those DTCs first.

D1 DRIVE CYCLE TEST

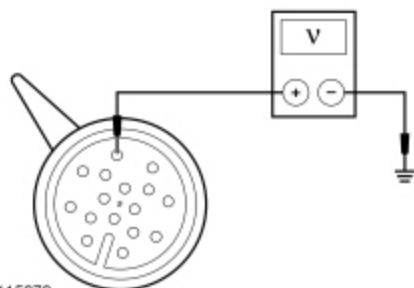
- Connect the scan tool.
- Enter the following diagnostic mode on the scan tool: TSS PID: TSS_SRC .
- Monitor the TSS_SRC PID while road testing the vehicle. Drive the vehicle so the transmission upshifts and downshifts through all the gears.

Does the TSS sensor rpm increase and decrease with engine rpm and vehicle speed?

Yes	REFER to Shift Point Road Test in this section for further diagnosis.
No	GO to D2 .

D2 CHECK TSS SENSOR SIGNAL CIRCUIT FOR POWER

- Ignition OFF.
- Disconnect: Transmission Vehicle Harness [C1575](#).
- Ignition ON.
- Measure the voltage between the transmission vehicle harness [C1575](#) Pin 1 and ground.

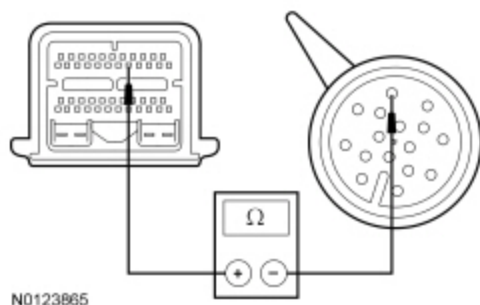


Is the voltage less than 4.8 volts?

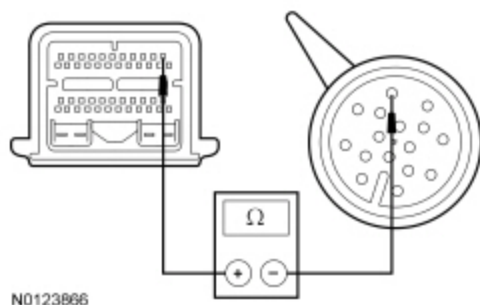
Yes	GO to D3 .
No	GO to D6 .

D3 CHECK THE TSS SENSOR SIGNAL CIRCUIT FOR AN OPEN

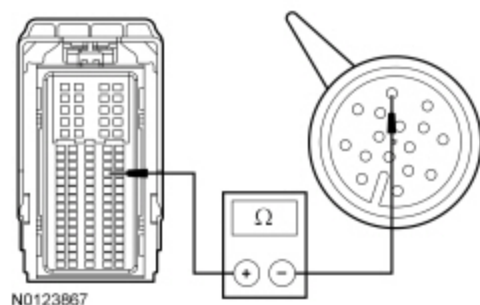
- Ignition OFF.
- Disconnect: PCM [C175T](#).
- Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.
- Vehicles equipped with a 3.7L engine, measure the resistance between the transmission vehicle harness [C1575](#) Pin 1 and PCM [C175T](#) Pin 16.



- Disconnect: PCM [C1381T](#).
- Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.
- Vehicles equipped with a 5.0L or 6.2L engine, measure the resistance between the transmission vehicle harness [C1575](#) Pin 1 and PCM [C1381T](#) Pin 1.



- Disconnect: PCM [C1551E](#).
- Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.
- Vehicles equipped with a 3.5L engine, measure the resistance between the transmission vehicle harness [C1575](#) Pin 1 and PCM [C1551E](#) Pin 27.

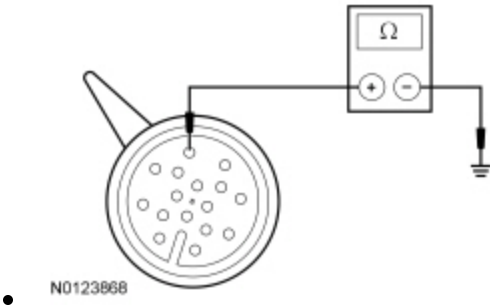


Is the resistance less than 5 ohms?

Yes	GO to D4 .
No	INSPECT and REPAIR transmission vehicle harness TSS signal circuit for an open. If an open circuit is not found, REPLACE the PCM, REFER to Section 303-14. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. TEST the system for normal operation.

D4 CHECK THE TSS SENSOR SIGNAL CIRCUIT FOR A SHORT TO GROUND

- Measure the resistance between the transmission vehicle harness [C1575](#) Pin 1 and ground.

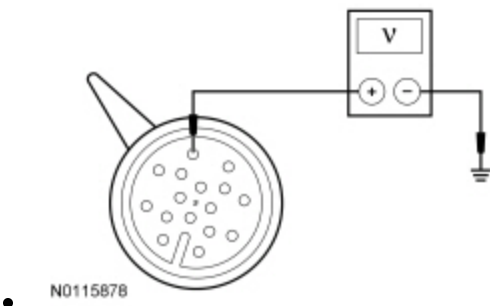


Is the resistance greater than 10,000 ohms?

Yes	GO to D5 .
No	INSPECT and REPAIR transmission vehicle harness TSS signal circuit for a short to ground. If a short to ground is not found, REPLACE the PCM, REFER to Section 303-14. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. TEST the system for normal operation.

D5 CHECK THE TSS SENSOR SIGNAL CIRCUIT FOR A SHORT TO POWER

- Connect: PCM [C175T](#).
- Connect: PCM [C1381T](#).
- Connect: PCM [C1551E](#).
- Ignition ON.
- Measure for voltage between the transmission vehicle harness [C1575](#) Pin 1, and ground.

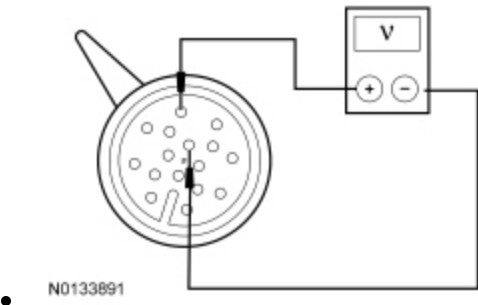


Is voltage greater than 5.1 volts?

Yes	INSPECT and REPAIR transmission vehicle harness TSS signal circuit for a short to power. If a short to power is not found, REPLACE the PCM, REFER to Section 303-14. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. TEST the system for normal operation.
No	GO to D6 .

D6 CHECK THE TSS SENSOR GROUND CIRCUIT

- Measure the voltage between the transmission vehicle harness [C1575](#) Pin 1 and [C1575](#) Pin 5.



Is voltage less than 4.8 volts?

Yes	REPAIR the transmission vehicle harness signal return circuit for an open. If an open is not found, REPLACE the PCM, REFER to Section 303-14. PROGRAM the PCM with the latest calibration level. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. TEST the system for normal operation.
No	REPLACE the molded leadframe, REFER to Shift Solenoids (SS) in this section. CLEAR the DTCs. TEST the system for normal operation.

PINPOINT TEST E : OSS SENSOR

NOTE: Refer to the Transmission Vehicle Harness Connector illustration within the Transmission Connector Layouts procedure in this section.

NOTE: Refer to the Transmission Leadframe Connector illustration within the Transmission Connector Layouts procedure in this section.

NOTE: If the scan tool is unable to access the transmission PIDs, refer to Powertrain Control/Emissions Diagnosis (PC/ED) manual.

NOTE: Always drive the vehicle in a safe manner according to driving conditions and obey all traffic laws.

NOTE: Read and record all DTCs.

NOTE: If DTCs P0705, P0706, P0707, P0708, P0709, P1702, P1705 and/or P1921 are set, diagnose those DTCs first.

E1 DRIVE CYCLE TEST

- Connect the scan tool.
- Enter the following diagnostic mode on the scan tool: OSS PID: OSS_SRC .

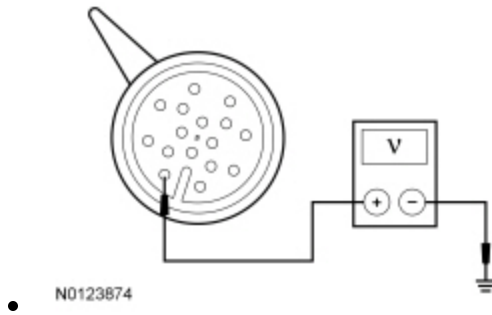
- Monitor the Output Shaft Speed (OSS) sensor PID while road testing the vehicle. Drive the vehicle so the transmission upshifts and downshifts through all the gears.

Does the OSS sensor rpm increase and decrease with engine rpm and vehicle speed?

Yes	REFER to Shift Point Road Test in this section for further diagnosis.
No	GO to E2 .

E2 CHECK OSS SENSOR SIGNAL CIRCUIT FOR POWER

- Ignition OFF.
- Disconnect: Transmission Vehicle Harness [C1575](#).
- Ignition ON.
- Measure the voltage between the transmission vehicle harness [C1575](#) Pin 15 and ground.

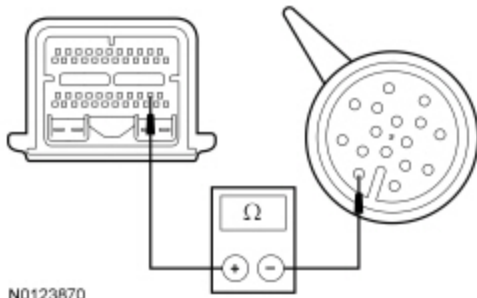


Is the voltage less than 4.8 volts?

Yes	GO to E3 .
No	GO to E6 .

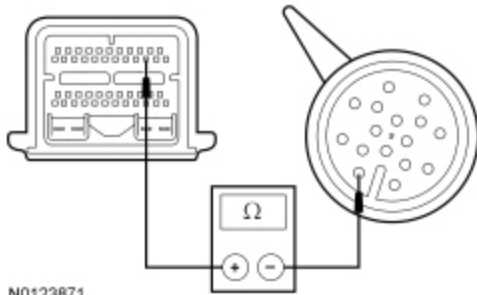
E3 CHECK THE OSS SENSOR SIGNAL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: PCM [C175T](#).
- Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.
- Vehicles equipped with a 3.7L engine, measure the resistance between the transmission vehicle harness [C1575](#) Pin 15 and PCM [C175T](#) Pin 25.



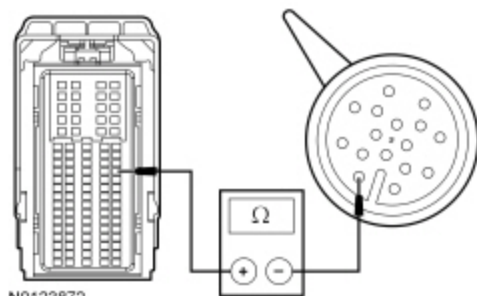
• N0123670

- Disconnect: PCM [C1381T](#).
- Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.
- Vehicles equipped with a 5.0L or 6.2L engine, measure the resistance between the transmission vehicle harness [C1575](#) Pin 15 and PCM [C1381T](#) Pin 14.



• N0123671

- Disconnect: PCM [C1551E](#).
- Inspect the connector for damaged or pushed out terminals, corrosion, loose wires and missing or damaged seals.
- Vehicles equipped with a 3.5L engine, measure the resistance between the transmission vehicle harness [C1575](#) Pin 15 and PCM [C1551E](#) Pin 12.



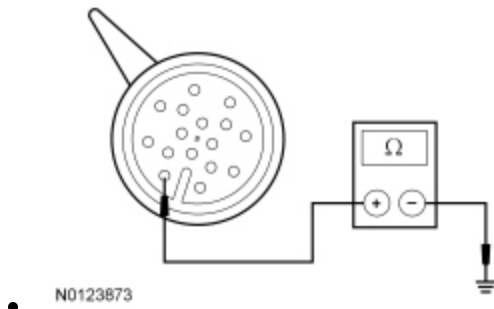
• N0123672

Is the resistance less than 5 ohms?

Yes	GO to E4 .
No	INSPECT and REPAIR transmission vehicle harness OSS signal circuit for an open. If an open circuit is not found, REPLACE the PCM, REFER to Section 303-14. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. TEST the system for normal operation.

E4 CHECK THE OSS SENSOR SIGNAL CIRCUIT FOR A SHORT TO GROUND

- Measure the resistance between the transmission vehicle harness [C1575](#) Pin 15 and ground.



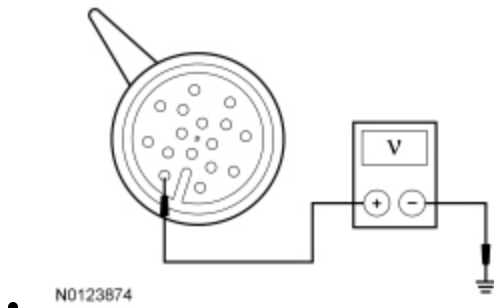
N0123873

Is the resistance greater than 10,000 ohms?

Yes	GO to E5 .
No	INSPECT and REPAIR transmission vehicle harness OSS signal circuit for a short to ground. If a short to ground is not found, REPLACE the PCM, REFER to Section 303-14. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. TEST the system for normal operation.

E5 CHECK THE OSS SENSOR SIGNAL CIRCUIT FOR A SHORT TO POWER

- Connect: PCM [C175T](#).
- Connect: PCM [C1381T](#).
- Connect: PCM [C1551E](#).
- Ignition ON.
- Measure for voltage between the transmission vehicle harness [C1575](#) Pin 15, and ground.



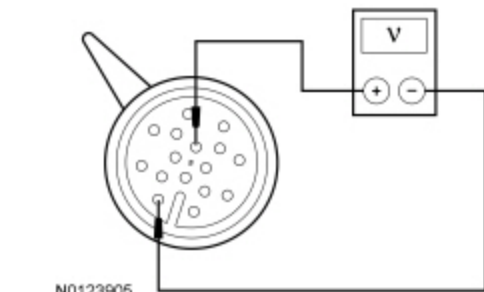
N0123874

Is voltage greater than 5.1 volts?

Yes	INSPECT and REPAIR transmission vehicle harness OSS signal circuit for a short to power. If a short to power is not found, INSTALL a new PCM, REFER to Section 303-14. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. TEST the system for normal operation.
No	GO to E6 .

E6 CHECK THE OSS SENSOR GROUND CIRCUIT

- Measure the voltage between the transmission vehicle harness [C1575](#) Pin 15 and [C1575](#) Pin 5.



Is voltage less than 4.8 volts?

Yes	REPAIR the transmission vehicle harness signal return circuit for an open. If an open is not found, REPLACE the PCM, REFER to Section 303-14. PROGRAM the PCM with the latest calibration level. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Solenoid Body Strategy in this section. TEST the system for normal operation.
No	REPLACE the molded leadframe, REFER to Main Control in this section. CLEAR the DTCs. TEST the system for normal operation.